



HK IMO Training 2021-2022
Problem Set 24
For 2 April 2022



Problem 93. Let n be a positive integer. In how many ways can we fill in the numbers $1, 2, \dots, 3n$ into the cells of a $3 \times n$ table such that the n numbers in each row are in ascending order (from left to right), and the 3 numbers in each column are in ascending order (from top to bottom)?

Problem 94. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x+y)f(y) = |x| f\left(y + \frac{y^2+1}{x}\right)$$

for any real numbers x and y with $x \neq 0$.

Problem 95. Let $n > 1$ be an integer. Prove that there exists an integer m with $\left\lfloor \frac{n}{2} \right\rfloor \leq m \leq n-1$, and integers $a_1, a_2, \dots, a_{n-m}$ such that $a_1 > 0$ and

$$\frac{a_1}{m+1} + \frac{a_2}{m+2} + \dots + \frac{a_{n-m}}{n} = \frac{1}{\text{lcm}(1, 2, \dots, n)}.$$

Problem 96. Let P be a point inside an acute $\triangle ABC$, and let Q be the isogonal conjugate of P with respect to $\triangle ABC$. Let D, E, F be the midpoints of the minor arcs BC, CA, AB of the circumcircle of $\triangle ABC$ respectively. The rays DQ, EQ, FQ meet the circumcircles of $\triangle BCP, \triangle CAP, \triangle ABP$ at X, Y, Z respectively. Prove that P, X, Y, Z are concyclic.